CREATURES OF CONTEXT

Understanding Behavior in a World We Share

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About this edition

This book grew from questions I wanted to understand better. AI contributed substantially to research, drafting, and rewriting. I chose the direction, questioned the results, and made editorial decisions. The book remains a reader beta; its account of the process accompanies the manuscript.

This book brings research into conversation with ordinary decisions and shared obligations. Its opening questions about threat, reward, and meaning are starting points for inquiry, not a diagnostic model. Imagined situations are identified as illustrations; the notes provide sources for empirical findings. Practical suggestions are proposals unless a tested intervention is specifically identified. Health discussions provide general education rather than individualized care.

The notes distinguish evidence, interpretation, and important limitations, including where only an abstract or a secondary account was consulted. This edition has received editorial review, not independent specialist review.

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INTRODUCTION

The person in the room

Imagine receiving a message from your manager: *Can we talk tomorrow?*

The message contains four words and almost no information about its purpose. Perhaps you have been expecting good news. Perhaps the last conversation ended with a warning. Perhaps your manager sends this message to everyone before discussing next week's timetable. The words remain the same. What you encounter when you read them does not.

You might keep eating dinner. You might put down your fork.

To explain the difference, we need more than the message. We need the relationship, the recent past, the stakes of the conversation, and something about the person opening it. We may need to know whether that person has slept, whether money is tight, and whether there is anyone with whom to share the uncertainty. These details help make the event.

From the outside, there is an anxious employee. From the inside, there may be rent, a previous humiliation, an ambiguous promise, and a whole evening in which nothing useful can be done.

This book begins with the thought that explanations should be at least as interesting as the people they explain.

Calling someone irrational can close a conversation precisely where the useful questions begin. Calling a response biological can have the same effect if it sounds like the end of the explanation. So can calling it cultural, learned, genetic, or a matter of choice. Each word points toward something worth understanding. None can do the work alone.

We are animals whose lives develop through relationships with a world. We inherit bodies with capacities and limits. We learn what happens when we reach, wait, speak, withdraw, trust, or refuse.

Tools extend what we can do. Institutions arrange what we must do. The body reading a message is already carrying a history.

Desmond Morris brought a zoological perspective to a wide readership in books including *The Human Animal*, published in 1994.[1] The question pursued here is what happens when we give the animal's changing circumstances as much attention as its inheritance. A language, a work schedule, and a relationship are parts of the world through which a human life develops. At the same time, a schedule cannot abolish the need for sleep, and a speech about resilience cannot remove pain.

What a better explanation is for

A missed deadline might call for clearer instructions, a smaller workload, treatment for a health problem, or a first imperfect attempt at feasible work. Those possibilities require different actions from different people. Calling all of them a failure of discipline leaves us with very little to do except demand more discipline.

The ambition of this book is to improve the match between explanation and action. Context includes the person's capacities and responses; it does not place every cause somewhere else. Understanding can also leave a disagreement intact. People may know what an arrangement costs and still differ over who should pay.

An explanation can help us prevent harm, arrange support, and decide what protection is needed. It cannot settle every moral question on our behalf.

The journey

We begin with three questions: what needs protecting, what draws us forward, and what makes something worth caring about? They give us a way into a decision that seems to pull in several directions. Following the answers will take us into bodily responses, learned expectations, relationships, and the work of living together.

The first part follows the science of responses and the histories and differences that shape them. The second moves into sleep, movement, emotion, and the difficulty of turning intentions into action. The final part begins with belonging, then follows commitments into disputes over shared work and the practices we hand on.

We will encounter experiments, observations, theories, and disagreements about what the evidence means. Their differences matter. So does the distance between an average finding and a particular life. The notes supply sources and qualifications; an attractive explanation should remain open to correction.

The aim is not perfect regulation. There are good reasons to be angry, restless, bereaved, excited, uncertain, or dissatisfied. Some of those states ask us to change something. Some ask us to endure something we cannot yet change. The wish to eliminate discomfort can obscure the difference.

The message is still on the phone. Before tomorrow’s conversation, we can at least become more careful about the story we tell around it.

PART I

Systems in context

A response becomes more intelligible when we can see what it is answering. These chapters begin with threat, reward, and meaning, then ask how experience and individual variation alter the picture.

CHAPTER 1

What matters right now

Consider an imagined job offer you want and are afraid to accept.

The salary is better. The work sounds interesting. The move would take you away from people who make your present life possible. You imagine succeeding, then imagine failing somewhere unfamiliar. You begin explaining your reasons to a friend and hear yourself making a case for both decisions.

Which response is the real one?

The question assumes that an offer should have one meaning. In fact, several things have changed at once. There is something to gain, something to lose, and a possible future whose value is still difficult to judge.

What needs protecting, what attracts us, and why does the commitment matter? These questions draw on different research traditions and work at different levels.[2] Here they give us somewhere to begin examining the offer.

What is at stake?

Start with the fear. Failure is the first answer, but failure can mean several things. Losing the job? Looking inexperienced? Moving the family and discovering that the new life does not fit? Becoming responsible for work without having the authority to do it well?

Those are different risks. A conversation about expectations could reduce one uncertainty. Savings might make another risk bearable. Training could help with unfamiliar work. Reassurance about being talented would not shorten a journey or replace the people who currently share care.

We should be equally careful about reading fear from an outward sign. Stillness could reflect attention, restraint, or exhaustion. A fast heartbeat does not identify the thought accompanying it.

Researchers distinguish reports of fear, defensive behavior, and bodily responses, while debating how these should be related in an explanation.[3] Naming a brain region would not tell us which part of the job offer is troubling this person.

Now consider the attraction. The money may matter because it would remove a recurring worry. The work may offer a skill the person wants to learn. Being selected may feel like recognition after years of being overlooked. All are possible rewards, but accepting the role would not necessarily supply all of them.

The offer is not a working day. Getting selected may feel wonderful; the job still has to be lived. It is possible to want the offer more than the work, and it is possible to want the work enough to accept days that are not especially pleasant. A task can be difficult and still be one we are glad to have undertaken.

Meaning enters there. The job may represent a promise made to oneself, a craft worth practicing, or a chance to serve people whose needs matter. It may also fit an inherited picture of success that has begun to feel less convincing.

Find something the explanation could get wrong

The person considering the offer now has two plausible accounts of the hesitation. Perhaps unfamiliar responsibility feels more threatening than it will prove to be. Perhaps the role really would remove time and support that the household cannot spare.

If the first account is right, more information about the work and available training could change the decision. If the second is right, a clearer description may make the problem more apparent rather than reassuring anyone. Calling both of them a lack of confidence would conceal the difference.

The applicant asks what an ordinary week contains. The answer reveals regular evening work that the advertisement did not mention. A visit also shows that much of the role involves supervising the activity they had hoped to practice themselves. The

salary remains attractive. Two other parts of the offer have changed under inspection.

The hesitation is beginning to look less mysterious. Some of it concerns a concrete conflict over time. Some concerns a mismatch between the work imagined and the work offered. The remaining anxiety about doing something unfamiliar may still call for courage or practice.

The visit has supplied information that another evening of weighing feelings could not. The applicant can ask current staff how the evening duties are shared and test the impression the visit has left. There is now a specific question to pursue.

Who can make the next change?

Suppose the applicant asks whether the evening duties can be shared or reduced. The employer declines. The person can accept the cost, negotiate elsewhere, look for another offer, or stay.

The household has a voice in the decision too. If the applicant’s partner would take on more unpaid work, that cost belongs inside the account of what the higher salary would buy.

The applicant declines the offer and asks for a smaller new responsibility in the current job. It provides some of the learning they wanted, but not the pay increase. They still feel disappointed when the original role is filled.

We will begin the closer scientific inquiry with a word that often arrives before any of this detail: stress. It can describe an impossible workload or the anticipation of a speech someone deeply wants to give. Before deciding how to reduce it, we need to know what it is hiding.

CHAPTER 2

What stress hides

The volunteer stands in front of a panel and gives a speech. The panel watches without encouragement. Then comes mental arithmetic, performed aloud while errors are corrected and the task begins again.

This is one of the more reliable ways to elicit a cortisol response from a healthy adult in a laboratory.

Researchers call this the Trier Social Stress Test. Its ingredients are ordinary enough: difficult work, public judgment, and very little control over the outcome. A synthesis of laboratory studies examined which demands produce this hormonal response. Social evaluation and limited control were a particularly potent combination, though these features also overlapped with effort and unpredictability.[4]

The volunteer is trying to perform while other people decide what that performance says about them.

The usual story about stress says that the body is ancient, modern life is confusing, and an email or audience can set off the same alarm as a predator. It is an appealing story because it makes a strong reaction feel less absurd. It also makes the body seem rather stupid.

People do not need to mistake a panel for teeth. Judgment can affect work, belonging, reputation, money, and the private confidence that one can do what is being asked.

The word *stress* can cover useful preparation, prolonged strain, and responses that no longer fit a situation. By itself, it does not tell us which of these is happening, or what would help.

One word, several events

During the Trier test, researchers can collect saliva and watch cortisol rise and fall. Cortisol participates in the body's ordinary work of adjusting energy and other processes. It also changes across the day, with waking, activity, illness, medication, and much else besides. A saliva sample tells us how one chemical response changed during one chosen stretch of time. It does not contain the experience in miniature.

The rest of the person has not stood still while the sample was collected. Heart activity may change. Attention may tighten around mistakes and signs of judgment. The volunteer may feel anxious, irritated, embarrassed, sharply focused, or determined to do well. They may want to escape, argue, try harder, or go blank.

A review of Trier-test studies found average changes in both cortisol and unpleasant emotion. Only six studies reported how changes in biological measures related to changes in mood, and their results were inconsistent.[5] A large hormonal response and severe felt distress need not occur in the same person.

Outside the laboratory, life is less considerate about holding its variables still. In studies that repeatedly sampled people during ordinary days, reported stress and changes in cortisol, heart rate, or blood pressure often failed to line up neatly. Walking upstairs, drinking coffee, sleeping badly, taking medicine, arguing with a partner, and worrying about tomorrow all arrive in the same body.[6]

The dream of a single reading is understandable. If someone has been told that their suffering is exaggerated, an objective number can look like a witness who will finally be believed. Yet the same number can be turned against them. The reading is low, so nothing serious is happening. The reading is high, so the person must be damaged. Neither conclusion follows.

We use *stress* for a fever, an argument, a hard workout, debt, grief, noise, public speaking, and the wait for a medical result. The shared word can obscure how differently we would need to respond to each.

Preparation can help - and still cost us

Preparation includes changes in circulation. Adrenaline and noradrenaline can raise heart rate and blood pressure, helping deliver blood to meet a demand. The same adjustment can serve a sudden challenge or the ordinary effort of climbing stairs. Sustained high blood pressure creates a different problem: over time, it can damage the cardiovascular system. A response's immediate usefulness and its long-term costs can therefore arise from the same activity.[7]

Another part of the response links the brain, pituitary gland, and adrenal glands. Signals along this route lead to the release of cortisol, which helps redirect energy resources. Cortisol also acts on cells that help regulate its release, reducing some of the signals that call for more of it. This is negative feedback: the response itself helps restrain further activity. The body has ways to end mobilization as well as begin it.[8]

Timing matters across these interacting systems. Repeated demands can call them into action again and again; a response can also persist after the demand ends. Sometimes too little of one response leaves other systems overactive. Bruce McEwen used *allostatic load* for the accumulating physiological costs of these patterns. The idea directs attention to regulation over time, rather than treating one unpleasant afternoon as a measure of damage.[7]

Once the volunteer leaves the silent panel, researchers can ask when cortisol returns toward its earlier level. They can also ask when attention becomes available for other things and whether the person would willingly face another audience.

Preparation does not have to feel good to be useful. A driver who sees a car move unexpectedly may need a faster heartbeat and a fierce narrowing of attention. The same narrowing can make a familiar name disappear just when a speaker needs it. A response can help with one part of a situation and make another part harder.

What does the response help the person do, and what does it begin to crowd out?

Sometimes the cost is immediate. Panic makes it hard to speak. Anger intended to defend against humiliation may ruin a conversation. Vigilance catches a real danger and makes sleep impossible afterward.

Sometimes the demand keeps returning. A caregiver remains responsible through the night. A worker has more tasks than time and no authority to change either. A tenant does not know whether the next notice will threaten their home. A family waits for a result that will decide what happens next. In these lives, alarm is not simply stuck after the danger has passed. The situation keeps asking for attention.

Over time, the cost can appear in several places. Sleep frays. Worry takes up room needed for other thought. Pain changes movement. People may drink more, eat differently, withdraw, or lose the activities that once helped them feel restored. Changes in hormones, the heart and blood vessels, immunity, and the nervous system may join that picture.[9]

Even *recovery* becomes slippery if we demand one biological score. After a laboratory challenge, researchers can calculate cortisol's return in several reasonable ways, and those calculations can disagree about the same response. Outside the lab, the thing we hope to get back is easier to recognize. Can the person sleep? Can they think about something else? Do appetite and ordinary energy return? Can their attention become available for someone they love? Most of all, has the situation stopped requiring vigilance?[10]

That last question is not under private control. Shift work, pain, insecure housing, unpaid care, discrimination, illness, and continuing danger can leave nowhere for the demand to end. A person can use every calming technique they know and still wake inside the same problem.

Settling enough to sleep, think, or avoid hurting someone can still matter enormously. At times it is the only reachable change.

What would reach the problem?

Imagine waiting for a medical result. Being told the exact hour of the phone call can end the uncertainty about when you will hear. Imagine receiving excellent feedback on work that still cannot fit into the available hours. The work has become clearer, not more possible. A friend saying “You’ve got this” may be kind. A friend picking up the children may change what the evening allows.

We often call all of these things *control*, but they are different. Knowing what will happen is predictability. Learning whether an action worked is feedback. Believing you can handle the situation is confidence. Perceived control is the belief that your actions can affect what happens; actual control is the capacity to do so. The belief and the capacity can come apart.

In an experiment with 168 healthy young adults, some participants could stop unpleasant noise and electrical stimulation; others could not. The working option reduced helplessness. Researchers also asked how much control participants thought they had. Those whose perceived control stayed low reported more helplessness than those with rising or moderate perceived control, even after accounting for which condition they had been assigned.[11]

The researchers also tried to increase confidence before uncontrollable stimulation, using positive feedback and memories of past competence. That exercise failed to raise measured self-efficacy or improve affect. It left the intended question unanswered: would a successful increase in confidence have helped? Actual options and beliefs about those options both belong in the inquiry.

Other people can sometimes provide the option one person lacks. They can share a night shift, make a call, lend money, give a ride, offer witness, use authority, or help someone leave. They can also judge, distract, make demands, or increase danger. The category *social support* is almost as broad as *stress*.

A review of experimental emotional-support studies found that its average stress-reduction estimate did not survive the authors’ main

adjustment for publication bias. A separate experiment with brief support from a stranger also found no general calming effect.[12] Those findings leave open what a trusted relationship can provide over time. Practical help raises other possibilities: a safer situation, less work, wider options, or a less lonely evening. What can this person help change?

The demand may also be built into a workplace, school, home, or public system. Asking an exhausted person to breathe through an impossible workload mistakes where the main leverage lies. Attempts to change working conditions face difficulties of their own. Workplace efforts to involve employees have produced uncertain average benefits, and some have added work to people who already had too much.[13] A committee can become another demand.

Room beyond the alarm

For the worker with too much to do, the inquiry can begin with the hours and the person who assigns the tasks. For the caregiver whose vigilance cannot end tonight, it may begin with someone who can take a turn. Sometimes the available change is partial: one night of rest, one duty shared, enough attention left for something besides the demand.

Those are outcomes worth choosing in advance. If the aim is a manageable workload, count the tasks and hours. If it is recovery, ask about sleep, energy, and time beyond the duty. The easiest thing to measure can otherwise become the goal by default.

A life widened beyond vigilance needs something to move toward. Yet pursuit has its own puzzles, including the possibility of continuing long after the pleasure has gone.

CHAPTER 3

Wanting more

Imagine that you have stopped enjoying the videos.

You are still watching. Nothing has forced the decision. There is simply another one, already beginning, and the next might be better. If asked whether this is what you want to do with the evening, you might say no while making the small movement that continues it.

Perhaps the watching is soothing, habitual, informative, or preferable to an unpleasant alternative. It exposes a puzzle: we can keep pursuing something while getting little pleasure from it.

The approach and the experience

In a 1989 experiment, Kent Berridge, Isabel Venier, and Terry Robinson studied rats after chemically damaging dopamine-producing neurons. Some stopped eating on their own. Had food ceased to be pleasant, or had something else gone wrong?

To find out, the researchers delivered taste solutions directly into the rats' mouths. A mirror beneath the chamber let a camera record the face and mouth. Sweet solutions still elicited the tongue movements and paw licking used to assess positive reactions to taste; bitter solutions still elicited aversive reactions. The broad pattern survived even in rats that would not feed themselves.[14]

The researchers were assessing observable taste reactions, a way to study hedonic impact in animals that cannot report their experience.

Researchers use distinct terms for components of reward. *Liking* describes hedonic impact, the pleasurable component of a reward. *Wanting*, in incentive-salience research, describes a process that makes a reward or its cues motivationally attractive. Learning concerns the relationships acquired through experience. These processes interact but can be separated experimentally.[15]

In ordinary life, anticipation and enjoyment often agree. Their disagreements catch our attention: the purchase that is exciting until it arrives, the reluctant conversation that becomes the best part of the day. Pursuit and pleasure are worth asking about separately.

Berridge and Robinson's incentive-sensitization theory addresses a particularly serious version of this separation in addiction. It proposes that, in susceptible individuals, repeated drug exposure can sensitize systems involved in cue-triggered wanting without a matching increase in pleasure. Animal and human research support aspects of the account, whose scope remains under investigation. Ordinary phone use does not establish this drug-related process.[16]

There is a further distinction between enjoying an activity and being glad to have done it. A difficult rehearsal may be frustrating from beginning to end and still feel worth attending. Pleasure is one reason to make room for something; it is not the only one.

What surprise can teach

Dopamine often appears in popular explanations as a quantity of pleasure delivered to the brain. That label is too crude to explain the research.

In a classic response pattern described by Wolfram Schultz, an unpredicted reward activates a dopamine neuron. Once the reward is fully predicted, its arrival produces no extra response. If the expected reward fails to arrive, activity drops below its usual level. As learning develops, the response appears at the cue that predicts the reward.[17]

The reward can stay the same while the signal changes. What matters to this response is the difference between what arrives and what was expected. Researchers call that difference a *reward prediction error*: information that can help update an expectation.

Dopamine has other roles, and its neurons do not all respond alike. A notification on a screen cannot tell us how much dopamine it has released or what the person looking at it feels.

The attraction of the chemical story is understandable. It seems to locate a difficult behavior in something concrete. Yet an explanation can sound more scientific while becoming less informative. “Dopamine made me do it” leaves unanswered what the person expected, which cue mattered, what alternatives were available, and why stopping was difficult on that occasion.

An environment with no last page

A printed book has physical stopping points. A streaming service can begin the next episode while you are deciding whether to continue. A shop has opening hours; an online shop can accompany you into bed.

Removing a pause, placing an invitation within reach, or making an unfinished possibility visible changes the decision a person encounters.

Imagine two versions of a video app. One asks you to choose a video, plays it, and ends. The other begins another video automatically and offers an unlimited sequence of alternatives. Both may contain material you value. They differ in how often stopping is made into an active task.

A pause could give viewers an occasion to notice the decision. Whether they welcomed it or used it to stop would have to be tested.

The consequences also depend on what the technology supplies. For someone who is isolated, an online community may be a valued source of companionship. For someone learning a skill, abundant demonstrations can be a gift. A tool can extend a capacity and make overuse easier within the same design.

Hunt Allcott and colleagues randomly assigned participants to deactivate Facebook for four weeks before the 2018 United States midterm election. Deactivation improved average subjective wellbeing and changed how participants used their time. It also reduced factual news knowledge. Time shifted into activities including television and time with family and friends.[18]

For other settings, the question is what a particular pattern of use contributes and what it displaces.

What is the activity doing for you?

Suppose you want to stop checking a shopping app.

Start with an occasion rather than a judgment about your personality. When did you open it? What were you expecting? What else was happening? Which part felt compelling: looking, choosing, buying, waiting for delivery, or imagining a different version of your life?

If browsing is a pleasant hobby within your budget, the problem may be smaller than a general rule about consumption suggests. If it repeatedly creates debt, secrecy, or distress, the consequences deserve attention regardless of whether the behavior fits a diagnostic category. If it fills the only unclaimed half hour in a demanding day, removing it without considering what the half hour needs to provide may leave an important question unanswered.

The activity may offer relief from worry or from a task that feels exposing. A purchase may promise membership, competence, beauty, or recognition. The product is sometimes only one part of what is being sought.

Consider which of these possibilities fits your experience. Wanting a better coat may simply mean needing a warmer one.

Judge a practical change against the cost that prompted it: interrupted sleep, unwanted spending, or a conversation repeatedly abandoned. Notice what is missed as well as what improves.

Curiosity, affection, ambition, play, and hope also draw us forward. A good environment should make room for pursuits we are glad to have followed.

The same promise will not attract everyone. Some people have learned that waiting pays; others have reasons to be skeptical. Some expect approach to bring interest and welcome; others have learned to watch for the cost.

To understand those differences, we have to go back to how a world becomes familiar.

CHAPTER 4

Learning what to expect

A child is offered a small reward now or a larger one later. The familiar interpretation asks whether the child can resist temptation.

Another question belongs in the room: does the child have reason to believe the offer?

Celeste Kidd, Holly Palmeri, and Richard Aslin began their experiment with an art project. An adult offered better supplies if the child waited. In one condition, she returned with them. In the other, she returned empty-handed. A second promise, this time about better stickers, was kept or broken in the same way.

Then came the marshmallow: one now, or two if the child waited for the adult to return. Among the twenty-eight children, those assigned to the reliable adult waited about twelve minutes on average, compared with about three minutes in the unreliable condition.[19] The treat presented the same choice, but the person making the promise now had a different record.

A later extension with sixty children in a familiar school setting found a smaller difference; its preregistered tests were inconclusive.[19] The original result gives us a reason to examine the adult's reliability when interpreting the child's wait.

What the marshmallow could not tell us

The waiting task became famous partly because early work linked longer delay with later outcomes. The idea is appealing: observe a small act of restraint, glimpse the adult to come.

Tyler Watts, Greg Duncan, and Haonan Quan revisited that relationship using a larger, more diverse sample and a modified version of the task. Associations with later achievement became substantially smaller after accounting for factors including family

background, early cognitive ability, and the home environment. Associations with later behavioral outcomes were weaker still.[20]

A preregistered analysis of the same longitudinal data by Laura Michaelson and Yuko Munakata reached a different conclusion. Children who waited the full seven minutes had fewer behavioral problems in adolescence, even after accounting for earlier behavioral problems. The researchers used different outcome measures and statistical adjustments.[21] The disagreement concerns what waiting predicts and how to explain the association; it remains open to scrutiny.

Kidd's experiment asks a different question: does changing an adult's reliability change waiting in the task? It does not test long-term prediction. The longitudinal analyses examine that prediction, but cannot tell us what would happen if children were trained to wait longer. A behavior can carry information about a child's future without being the cause of that future.

That distinction reaches beyond marshmallows. Schools, families, and institutions often need to assess behavior. The risk comes when a useful observation becomes a total explanation. A child who struggles to wait, ask for help, settle into work, or trust an adult may have something to learn. The adult may have something to change.

Repeated experience becomes an expectation

Imagine two classrooms.

In one, asking for clarification usually produces help. In the other, the response is unpredictable: help one day, public irritation the next. It would be unsurprising if children became more willing to ask questions in the first room. Yet if we observed only a quiet child in the second, we might call the behavior a lack of curiosity.

The comparison suggests a question to ask the child: what happened the last time you spoke?

Learning is rarely delivered as a clean verbal lesson. People learn from what follows an action, from what others do, from repeated

patterns, and from the explanations available to them. A rule printed on a wall can be contradicted by what happens when somebody tests it.

A family may sincerely say that mistakes are acceptable. If each mistake repeatedly leads to a long, frightening confrontation, the words and the experience offer different information. Understanding an expectation requires looking at both the declared rule and the consequences.

These examples also show why the past cannot be neatly sealed away. A person may carry an old expectation into a new setting, but a new setting may continue to confirm it. Before asking someone to stop anticipating criticism, it is worth checking whether criticism has actually become less punishing.

Different experiences can leave different demands

The phrase *adverse childhood experience* covers events that should not be treated as interchangeable.

Katie McLaughlin and colleagues proposed separating exposure to threat from deprivation of expected cognitive and social experience. Their framework asks both what a child repeatedly encounters and what opportunities to learn are missing.[22]

Experience helps shape connections between neurons. Some connections strengthen; others are removed as development proceeds. Drawing partly on animal studies of sensory deprivation, the researchers proposed that severely restricted cognitive and social experience could alter this process in brain regions supporting complex thought. Language and responsive interaction supply activity through which capacities develop. Their absence may affect development through more than an ongoing stress response. The specific pathway from broad social deprivation to changes in human neural connections remains a hypothesis in this account, not something a brain scan directly reveals.[22]

Threat raises another question: which signals become especially important to recognize? In a study by Seth Pollak and Pawan Sinha,

children watched scrambled photographs gradually resolve into facial expressions. At intervals they identified the emotion and rated their confidence. Children with histories of physical abuse identified angry expressions from less visual information than the comparison group. They needed more information for sadness; the groups did not differ significantly for fear or happiness.[23]

The abused children were recognizing anger earlier, rather than simply choosing the anger label more often when wrong. The authors proposed that children facing physical threat might learn to recognize anger from fewer visual clues.[23]

A child can face threat and deprivation together. Experiences differ in timing, duration, and intensity, and people bring other capacities and relationships to them. Counting difficult events can reveal a population pattern while leaving the process in a particular life largely unexplained.

An adult’s habit of avoiding conflict does not, by itself, reveal what happened in childhood. The same outward behavior can have different histories. Reconstructing a person’s past from a present-day trait risks turning understanding into invention.

A learned response may fit an earlier environment better than the present one. It may also have carried costs from the beginning: recognizing danger earlier could help a child anticipate a confrontation while leaving the harm intact.

Change is possible without being simple

A body and a history are not reset by entering a better room.

If trust has repeatedly been costly, one reassurance may not provide much evidence. The new person may be sincere; the old expectation may still be understandable. Change has to include what happens next.

Research on children exposed to severe institutional deprivation provides one unusually strong form of evidence that changing care can affect development. In the Bucharest Early Intervention Project, children already living in institutions were randomly assigned to an

intervention providing foster care or to care as usual. Follow-up research reported benefits for cognitive development, with continuing variation in outcomes and the importance of later placement circumstances.[24]

The intervention showed that changing care could alter development after severe deprivation. Its extreme circumstances limit comparisons with ordinary parenting.

For a reader trying to alter an established pattern, a practical question is what new experience would be informative. If the expectation is that asking for help brings humiliation, repeatedly receiving respectful help may matter more than repeating a sentence about confidence. If the expectation concerns real material insecurity, reliable access to resources matters. The word *safety* has little explanatory value until we say what makes a situation safer.

Where harm continues, protection takes priority over asking someone to reinterpret the experience.

Care without a perfect childhood

Parents are often offered explanations that make them responsible for everything and guidance that makes success impossible to recognize.

A child is growing among siblings, peers, schools, neighborhoods, institutions, bodily differences, and events no caregiver controls. Caregiving takes place among these influences, with help or interference from them.

Consider a parent who regularly returns home exhausted and finds a child’s demands difficult to tolerate. There may be useful changes in how conflict is handled. There may also be a need for sleep, practical help, money, or fewer demands. Advice that attends only to the interaction can miss what repeatedly makes the interaction difficult.

Preventing hurtful behavior may require both better ways of handling conflict and practical support for the caregiver.

Dependability is a more workable aim than a perfect childhood: words and actions that increasingly agree, help that can be sought, and repair when something goes wrong. The adults need enough support to offer that dependability more than occasionally.

People with similar histories can develop differently. Even those who share a home have different experiences. A new opportunity may be easy for one person to use and difficult for another. Those differences deserve a closer look.

CHAPTER 5

Averages do not live your life

Imagine someone who talks freely over dinner with close friends, then says almost nothing in a meeting the next morning. A friend calls them outgoing. A colleague calls them quiet. Both have evidence.

We could choose one description and explain away the other. Or we could ask what changed between the dinner table and the meeting: the company, the subject, the right to interrupt, the cost of saying something foolish. A trait can summarize a person while leaving a particular silence unexplained.

A person contains variation too

William Fleeson repeatedly asked people about their personality-related states during ordinary life. Three studies, each lasting two or three weeks, found substantial variation within individuals alongside stable differences in their average patterns.[25]

Imagine keeping each person’s observations together rather than replacing them immediately with one score. Some occasions would be quiet, others sociable. Two people’s records could overlap considerably, with sociable occasions more frequent in one of them.

Fleeson describes traits as distributions of states: records of how often different states occur. Their averages summarize patterns without fixing a person permanently at the center. The pattern alone cannot tell us how much comes from capacities, preferences, or the situations the person repeatedly encounters.

This way of looking also changes a self-description. “I am bad at starting things” might conceal a pattern: perhaps starting is difficult when instructions are ambiguous, when the work will be judged

publicly, or when there is no clear first action. In other circumstances, the same person may begin readily.

What genes do and do not tell us

Genetic differences contribute to variation in many human traits. Tinca Polderman and colleagues' large synthesis of twin research documents the breadth of that contribution across the traits studied.[26]

But *heritable* is easily misread as *unchangeable*.

Heritability describes variation within a studied population under particular conditions. It is not the percentage of an individual's trait caused by genes. Its value can differ when the population or environments differ, and a high estimate does not tell us that an environmental intervention would be ineffective.[27]

Consider height. Among children with similarly adequate nutrition, differences in food supply may explain little of the variation in height. A severe food shortage would still affect their growth. A resource can be essential even when its availability varies too little to explain much of the difference between people.[27]

The practical question is how much a helpful change can accomplish. A quieter room may make concentration easier. Other difficulties persist even after noise, interruptions, and unclear demands have been addressed.

Sensitivity is not a rank

People also differ in how strongly they respond to aspects of their surroundings.

Michael Pluess's framework of environmental sensitivity proposes that people who are more affected by adverse conditions may also benefit more from supportive ones.[28] In that account, sensitivity includes responsiveness to opportunity. It is a proposal about differences in response, with continuing questions about how reliably to identify them.

Turning it into fixed personality types would obscure the question worth asking: sensitive to what?

A person may find one kind of stimulation difficult and another welcome. Noise, unpredictability, criticism, social density, physical exertion, and novelty are not a single substance delivered at different strengths. Describing which conditions matter is more informative than saying that someone is sensitive to life.

The setting has assumptions of its own. If a workplace provides only one way to concentrate, communicate, or recover, a difficulty attributed to an individual may partly reflect the narrowness of the available options.

Jobs, relationships, and shared spaces require coordination. Some demands follow from the work itself; others are conventions that could be changed.

When a difficulty persists

Consider an imagined person who repeatedly spends shared money on impulse. They have clear agreements, a supportive partner, and a reasonably stable life. The purchases continue, putting bills at risk. Looking for the next environmental adjustment can itself become a way to postpone facing how persistent the difficulty is. The partner needs protection from its costs as well as an explanation.

Persistence alone does not reveal a genetic cause. It does change the practical ambition: the person may need continuing help and agreed safeguards for shared money, rather than expect one change to remove the urge. A review of interventions, many of them clinical, found measured changes in personality traits, particularly emotional stability, across varied designs.[29] Those group findings leave open how far any one person’s difficulty will change.

Developing a capacity and limiting the cost of a difficulty are both worthwhile aims. Practice, treatment, and accommodations can serve different parts of that work.

For a less costly difficulty, consider someone overwhelmed by an unstructured networking event who enjoys conversation during a shared task. They could practice entering unfamiliar conversations and choose gatherings organized around an activity. Participation becomes easier to sustain when the setting offers a way in.

The person’s opportunities over time give us something to judge. An arrangement that keeps valued activities possible is doing useful work; a succession of restrictions that removes them calls for reconsideration.

Notice where the pattern holds

The dinner table and the meeting offer different views of the person we met at the start. Looking closely at when someone speaks can reveal a capacity that a single description misses. Looking across settings can also reveal how steadily a difficulty follows them.

Both patterns are informative. If concentration returns when interruptions stop, the interruptions deserve attention. If a difficulty survives several helpful changes, its persistence deserves attention too. The more detailed inquiry at the end of this book begins with observations like these.

The same care is needed with everyday advice. A real biological need can be shared while the opportunities to meet it differ. At the end of a long day, knowing that sleep matters may be the easiest part of getting it.

PART II

Everyday environments

These chapters follow sleep, movement, feelings, and action into the arrangements of an ordinary day. Sometimes a change makes the task easier. Sometimes it reveals a different task.

CHAPTER 6

A place for sleep

Trying hard is an odd way to fall asleep.

You can turn off a light, put down a book, and decide that work is finished. You cannot carry out sleep with the same obedience. Lying awake, you calculate how many hours remain before morning. The arithmetic is correct. It is also becoming another thing to do in the dark.

Well before that calculation, however, there was a decision about when the evening would end. Sometimes that is where the more revealing story begins.

Take an invented household. At half past ten, a parent finally sits down with a phone. The children are asleep. The kitchen is usable again. There is school paperwork to finish, but it can wait until morning. For the first time since breakfast, nobody is asking for anything.

The parent has already decided to go to bed earlier. Now the quiet has arrived, and giving it up feels like ending the day without having had a part in it. A funny message from a friend becomes a conversation. There is time for one more thing, then another. The alarm will ring at the same hour regardless.

Stopping would mean surrendering the first part of the day that felt chosen.

Two kinds of time

Sleep researchers distinguish the pressure that builds with wakefulness from the daily timing of sleep and alertness. In Alexander Borbély and colleagues' two-process model, the first rises while we are awake and declines during sleep. The second follows a circadian rhythm: an approximately daily cycle that changes our readiness for sleep. Together, they help explain why hours awake alone cannot tell us when someone will fall asleep.[30]

A nap illustrates the interaction. Sleep during the day reduces some of the accumulated pressure while the body's daily rhythm continues. Later, the same clock time meets a different history of wakefulness.

The surroundings enter this biology. In a small experiment, Anne-Marie Chang and colleagues had twelve healthy young adults read for about four hours before bed, for five evenings on a light-emitting tablet and five with printed books. Everyone completed both conditions, in a randomly assigned order. The room was dim and the tablet at maximum brightness. After the tablet condition, participants' melatonin rhythm was later, they took longer to fall asleep, and they felt sleepier the next morning.[31]

This prolonged, bright exposure gives us a concrete reason to take evening light seriously. The experiment did not separate light's color from its intensity or establish what a brief screen check would do.

Back in our imagined household, the phone also contains a friend.

Suppose the parent tries leaving the phone to charge downstairs. On the first evening, that removes the easy continuation of the conversation. On another, the parent stays up with a printed book instead. There is pleasure in the book, and reluctance to close it. The first attempt has made one explanation less convincing: the phone's presence cannot be the whole problem.

An hour that belongs to somebody

The most obvious solution is to move free time earlier. In this household, that means moving something else.

The other parent is also tired. Taking over the kitchen every evening would give one person an earlier break by postponing the other's. The school paperwork still needs doing. Tomorrow's meals will still need to be made. Even a modest change raises a question that a bedtime reminder cannot answer: whose obligation can move, and where will it go?

The parents try a smaller arrangement. On alternate evenings, one takes the remaining kitchen work while the other gets half an hour alone. They agree to leave the clean laundry unfolded. That means living with the basket in the bedroom for another day, which bothers one of them more than the other.

The first week goes unevenly. On one evening the protected time is used for the school paperwork. On another, a child needs attention. On a third, the parent reads for pleasure before ten and closes the book at the end of a chapter. The revised arrangement has created some room. It has also exposed a habit of treating unclaimed time as space for another obligation.

The next decision is more specific than "be disciplined." Can the paperwork be dealt with during the shared work period? When an interruption consumes someone's break, is there another opportunity? Does a finished chapter provide an ending the parent is willing to keep?

With the book open, the parent still has to decide when to close it. Tonight's pleasure may be worth tomorrow's cost; keeping to the plan may take practice. Sleep now has a place in the evening alongside some time the parent has chosen.

This household has another adult available and some discretion over the evening. A parent caring alone may have neither. Someone whose shift changes without notice cannot privately manufacture a predictable week. In those circumstances, advice to negotiate a better bedtime can become another demand placed on the person

with the least power to meet it. The practical question reaches whoever controls the schedule, the care, or the resources.

And the cost of short sleep needs more than a guess based on how familiar it feels.

In a laboratory study, Hans Van Dongen and colleagues randomly assigned healthy adults to four, six, or eight hours in bed each night for fourteen nights. During waking hours, they repeatedly tested performance and asked participants how sleepy they felt. One test required quick responses to signals appearing at unpredictable intervals. In the four- and six-hour groups, lapses accumulated across the days, while reported sleepiness rose initially and then changed much less.[32]

The two records told different stories. Feeling accustomed to a schedule was poor assurance that performance had stopped deteriorating. Familiarity had concealed an accumulating cost.

Our parent's assurance that "I can get through tomorrow" deserves another look. The household is deciding how to share a cost whose accumulation may be hard to notice.

When there is room, but sleep will not come

There is another possible continuation of the parent's story. The work is shared, some chosen time happens earlier, and the parent goes to bed. Then comes the arithmetic in the dark.

Even with time available, the parent lies awake. The difficulty now calls for a closer look at sleep itself.

For adults with chronic insomnia disorder, the American Academy of Sleep Medicine strongly recommends multicomponent cognitive behavioral therapy for insomnia, or CBT-I. It conditionally recommends against sleep hygiene as the only treatment.[33] CBT-I is a structured treatment that goes beyond general advice about routines and surroundings.

Persistent sleep difficulty with daytime impairment deserves clinical assessment, including consideration of other sleep disorders and health conditions.[34]

Making room for sleep still leaves room for an absorbing book, a conversation that runs late, and the ordinary untidiness of living together. An evening can be worth having even when it goes differently from the plan.

At half past ten, the clean laundry can remain in its basket. It will need attention tomorrow. The people who left it there need attention tonight.

CHAPTER 7

Room to move

Try to remember every time you moved yesterday. The walk to work may come readily to mind. Trips between rooms, time spent cooking, and the journey back for something forgotten are harder to reconstruct.

One way researchers study activity is to ask people to wear a device that records movement. Ulf Ekelund and colleagues brought together studies using these records and following participants over time. Greater recorded activity was associated with lower mortality risk.[35] The records captured movement people might overlook when asked how much they exercised.

They also captured lives already in progress. Illness could reduce movement, and other differences between more and less active people could contribute to the association. Bringing movement into an ordinary day raises a further question: where does it fit?

The World Health Organization’s guidance includes movement in transport, work, household life, and recreation. Planned exercise is one part of this larger category, and some activity counts even below the recommended amounts.[36] A useful possibility might be present in an ordinary journey, without adding a separate session to the day.

The route on Tuesday

An office worker in an imagined town decides to walk to a nearby park after work. Monday’s late meeting uses the daylight. On Tuesday, the worker reaches a fast road and turns back. The alternative entrance adds more time than a lunch break allows. The park is close on the map and difficult to reach.

Leaving shoes beside the desk helps with departure. A safe crossing and a break long enough to use it require decisions by planners and the employer. The worker might find another route; people who follow will still meet the same road.

The body making the journey

Even a usable route meets a particular body. Pain can make the return harder than the outward walk. Someone whose work is physically exhausting may need rest and safer work more urgently than another exercise target. A wheelchair user may need an accessible surface and an activity they enjoy. These are decisions about what movement to make possible, how much, and at what cost.

For depression, two reviews of randomized trials found possible benefits from exercise, while judging important parts of the evidence to be low certainty or limited by bias.[37][38] Those findings support discussing suitable activity as one part of care, including how the person would reach it, find support, and continue through a difficult week.

For some conditions, generic encouragement to do more is inappropriate. NICE's guidance on ME/CFS advises against programs based on fixed incremental increases in activity and against treating exercise as a cure. It calls for an individualized approach within the person's energy limits and appropriate specialist support.[39] Post-exertional worsening should not be interpreted as a failure of motivation.

A reason to return

Reaching a place is only the beginning of an outing. Someone who associates exercise with being mocked at school may be managing public attention as well as unfamiliar movements. A trusted companion or an activity without public comparison offers a different experience. Someone who dislikes a gym may welcome a dance floor, a garden, or a game.

There is pleasure in going somewhere, learning a movement, and becoming able to do something that was difficult before. Adapted activities can widen those possibilities.

A body can move because the music is good or because a game has begun.

CHAPTER 8

What to do with a feeling

Suppose a friend cancels an arrangement that mattered to you. The first explanation arrives quickly: you are unimportant to them. Anger follows, along with a message you could send immediately.

The friend could be ill, overwhelmed, careless, or avoiding you. You do not yet know. The alternatives offer something to ask about, but the evening has already been lost.

The message is still unsent. You reread the accusation and think about the reply it is likely to bring.

What should the message accomplish?

Do you want an explanation, an apology, a new arrangement, or a chance to say that the cancellation hurt? Those aims might overlap. Reading the draft with one of them in mind can reveal how little the accusation asks the friend to explain.

Emotion regulation is often imagined as a struggle at the end of a sequence: the feeling has arrived, and a person must stop it from showing. James Gross's research framework opens the picture earlier. People can influence the situations they enter, aspects of those situations, their attention, their interpretations, and their responses. His later account also asks how people recognize a need to regulate, select an approach, and carry it out.[40]

By the time this message is being drafted, the cancellation has happened. Some earlier possibilities are closed. Attention and interpretation remain available, along with the decision about what to send. On another occasion, a clearer agreement about when to confirm plans might change the situation before either person becomes upset.

Feeling better is one possible aim. You might also allow sadness to remain while listening to a grieving friend, or restrain amusement to avoid humiliating someone.

Reappraisal means changing how a situation is understood. The thought that your friend may be struggling could soften the initial interpretation enough to make a question possible. If this is the fifth cancelled plan, however, the question also needs to concern what can be relied upon. A generous explanation of each separate evening could keep postponing that conversation.

One observational study examined reappraisal ability in relation to stress and depressive symptoms. At high stress severity, stronger reappraisal ability was associated with fewer depressive symptoms when stressors were relatively uncontrollable, but with more symptoms when they were relatively controllable.[41] It leaves a causal question open: what would happen if people were taught to reappraise those situations?

For the cancelled plan, a more generous interpretation could open a conversation or help avoid one. The friend might have a good explanation tonight and still be someone with whom future arrangements need to change.

A feeling without a second trial

Before sending anything, you notice another thought: being this hurt is embarrassing.

There is the original feeling, and now a judgment about having it. Envy may be followed by shame at being envious. Anxiety may be followed by anger at failing to be calm. Sadness may become evidence that recovery is taking too long. The person has an emotion and a case to prosecute against themselves.

In laboratory, diary, and longitudinal research, people who habitually accepted their thoughts and emotions tended to show better psychological health, with patterns consistent with reduced negative reactions to stress.[42]

Acceptance, as used here, means allowing the experience to be present without treating its presence as a personal offense. You can acknowledge jealousy and decline to inspect a partner’s private messages. You can acknowledge anger and leave a conversation before doing harm.

The cancelled evening mattered. You can say so while asking what happened. Hurt can remain during that exchange.

Expression has an audience

A trusted listener, a hostile supervisor, a child, and a public audience have different responsibilities, capacities, and power. The same disclosure can invite care, confuse somebody who cannot help, or expose a person to consequences they have reason to avoid.

Research by Emily Butler, Tiane Lee, and James Gross complicates a universal rule against suppressing emotional expression. Some negative social effects of suppression were reduced among participants who endorsed more Asian cultural values, measured individually in the study.[43] Restraint takes its meaning partly from what the people present expect of one another.

The draft to the friend could become a private question or a public accusation. Posting it where mutual friends can see would bring a new audience into the dispute before the friend has answered. Silence has consequences too: if it becomes indefinite, the friend may never learn what the cancellation cost.

In a relationship where disagreement is safe, a pause can preserve the possibility of an honest exchange. Waiting until tomorrow may give you time to write what you mean. Repeatedly postponing the conversation would leave the grievance untouched. In a coercive relationship, direct disclosure can carry risks that an ordinary communication script cannot resolve; safety takes priority.

Suppose the next message asks what happened and says the evening mattered. It leaves a question the friend can answer. The disappointment remains, and the future of their plans is still open. For now, the message says something the accusation would have obscured: you wanted to spend the evening together.

CHAPTER 9

Beginning, repeating, avoiding

The document is open, time has been set aside, and the work is waiting. You adjust the heading, answer a nonurgent message, and search for one more example. Each action has a plausible relationship to work. None reaches the paragraph that needs to be written.

Take an imagined report writer in that position. The deadline is Friday. The report must recommend whether a project should continue, and the writer has figures showing that participation has risen. There seems to be enough to begin.

Another reminder of the deadline produces more alarm and another visit to the heading. Eventually the writer leaves the formatting alone and writes a rough paragraph.

What the first paragraph reveals

It recommends continuing the project because participation has risen. On the page, the gap is suddenly obvious: more people attending does not establish that the project helped them. The information on hand was enough to begin, but not enough to support the conclusion.

The writer marks the missing evidence and sends a colleague a specific question about outcomes. The paragraph is still poor. It has done something the rearranged heading could not: revealed what the report needs next.

Avoiding the paragraph postponed an encounter with that problem. Writing it exposed a genuine evidential need. Having found the need, the writer now has a good reason to wait for an answer. The document remains unfinished, but the reason has changed.

Procrastination is usually defined around unnecessary postponement despite expecting the delay to leave one worse off. Waiting for information, choosing a higher priority, and recognizing that work exceeds one's capacity can be sensible decisions.[44] The distinction takes knowledge of the task, not just a look at the calendar.

Piers Steel's synthesis found procrastination associated with several factors, including task aversiveness, delayed outcomes, beliefs about ability, and impulsiveness. His theoretical account combines how valuable and achievable a task seems with the timing of its consequences.[44]

Fuschia Sirois and Timothy Pychyl emphasize short-term mood regulation. In their account, turning away from an aversive task can improve how a person feels now while leaving the consequences to a future self.[45] For our writer, the easy message offers relief from uncertainty. Friday is still approaching, but the paragraph no longer has to be faced this minute.

The phrase *finish the report* hides several operations: deciding what it should say, finding evidence, drafting, checking an argument, and formatting the result. Some can become refuges from the others. Rearranging a folder may be necessary once and a refuge on the fifth return.

For feasible work repeatedly avoided, a rough attempt can reveal which operation is difficult. The attempt also needs somewhere to happen: a manager who punishes every imperfect proposal gives a writer a reason to conceal uncertainty. A private draft or an agreed exploratory stage can make it possible to expose a problem before being expected to solve it.

Opening the document can become routine. Deciding what belongs in it remains the writer's work.

What calls up an action?

Reaching for a phone on entering a room presents almost the reverse problem. The movement is familiar enough to escape notice until the screen is already lit. Here an action arrives readily, before much consideration of whether it is wanted.

Wendy Wood and Dennis Rünger describe habits in relation to repeated responses in recurring contexts. Habits and deliberate goals can work together: a person may intentionally arrange an activity that later becomes easier to initiate through familiar cues.[46]

Many valued activities would become cumbersome if every step required a fresh debate. Preparing food, caring for someone, and practicing a skill contain mixtures of routine and attention. A habit reduces the need to deliberate. That ease can also outlast the reason for wanting the action.

Frequent behavior alone tells us little about this process. A person can make the same considered choice every day, or repeat a task because a supervisor requires it. What matters is how the action is guided.

Habit formation is often sold with a deadline: repeat an activity for a certain number of days and it will supposedly become part of who you are.

Phillippa Lally and colleagues followed ninety-six volunteers who chose a daily eating, drinking, or activity behavior. Participants reported how automatically it occurred over twelve weeks. For the 39 participants whose records fitted the researchers' model well, the estimated time to approach their eventual level of automaticity ranged from 18 to 254 days.[47] The longer estimates extended beyond the observation period; they came from fitted curves, not from watching someone cross a finish line on day 254.

The spread gives little support to a universal deadline. It also concerns simple daily behaviors: drinking a glass of water makes different demands from writing a report or listening carefully to another person.

A plan can help before anything becomes automatic. Research on implementation intentions examines linking a recognizable situation with a planned response. Peter Gollwitzer and Paschal Sheeran's synthesis found benefits for goal attainment across studies of such if-then plans.[48] A deliberate plan specifies when to act; repetition may then help the action become easier to begin.

Suppose a reader puts a book beside the chair where they sit after lunch and plans to read a few pages before opening another application. On the first two days, the book is there when the meal ends. On the third, lunch is interrupted. The bookmark still marks a place to return to at the next lunch.

When a pause helps

An unwanted routine may benefit from making a response less immediately available. Moving an application off the first screen or turning off a nonessential notification can remove an invitation to check. The useful thing to watch is whether checking changes and what, if anything, is missed.

Work messages show why removing a cue may accomplish little. An employer who expects immediate evening answers and punishes silence has supplied a continuing reason to check. Even without that pressure, an application may offer reassurance, novelty, or contact that the person still wants. Removing the invitation leaves the question of what it supplied.

Some actions deserve more deliberation: spending money, sharing an accusation, agreeing to more work. A pause gives the person an opportunity to reconsider. Ease is useful when it serves a chosen activity; friction is useful when it interrupts an action before an unwanted consequence.

For the report writer, however, another pause before beginning would have prolonged the original difficulty. Writing brought the missing evidence into view. The helpful arrangement depends on which action needs to become easier and which needs to be interrupted.

Returning and repairing

The colleague replies that outcomes were never systematically recorded. The writer can now explain the gap and revise the recommendation, perhaps proposing a limited continuation that includes a way to assess benefit. Better evidence will require work beyond finishing the report. The draft now has a purpose: a decision someone can examine.

If the delay has created work for somebody else, repair includes an honest update and a feasible next commitment. A note naming the next action can make returning easier. An impossible recovery schedule creates another promise likely to be missed.

Psychological treatments have also been tested. A review of randomized trials found small average benefits for procrastination, with considerable variation; a small subset of cognitive behavioral therapy studies was promising but limited.[49]

The report still needs writing. What has changed is the work the writer can name: explain what the attendance figures establish, identify what remains unknown, and make a recommendation that acknowledges both. The heading can wait.

PART III

Shared worlds

Being known, having something worth doing, and sharing a workable set of rules are different goods. They can support one another and come into conflict. The final part begins with a place among other people, then asks what our commitments demand and what those who follow us will inherit.

CHAPTER 10

A place among others

A crowded calendar can contain very little companionship.

Consider someone whose day is full of necessary exchanges: a greeting at the door, questions at work, messages about schedules, a transaction on the way home. They speak to many people. They may still have nobody with whom they can be unfinished, uncertain, or quietly glad.

Another person spends much of the same day alone and enjoys it, expecting a conversation with a friend in the evening. Counting encounters would not tell us which person feels connected.

Loneliness concerns an unwelcome gap between the relationships a person has and the connection they want. Social isolation concerns the relative absence of contact or ties. Solitude can be wanted and restorative. These overlap in lived experience, but the distinctions keep us from prescribing company whenever we see aloneness, or assuming that company has resolved loneliness.[50]

What a health finding can tell us

Research linking social relationships and health has made loneliness a public concern. Two research syntheses found associations between loneliness, social isolation, living alone, and mortality risk, drawing on prospective studies that followed people over time.[50]

The studies are observational: health can affect relationships, and both are shaped by other circumstances. The associations help establish a public-health concern; they cannot supply an individual forecast.

Which connection is missing?

The word lonely can cover different absences.

A person may miss an intimate confidant while having many acquaintances. They may have a loving partner but no wider community. They may feel culturally out of place in a friendly room. They may have lost the particular person whose recognition made a day feel complete. Another social event will not necessarily answer each of these losses.

Before proposing a solution, it helps to make the absence more specific. Is the difficulty a lack of opportunities, a lack of safety, a lack of continuity, or the sense that nobody knows anything that matters about you? Is it recent, tied to a transition, or part of a longer pattern? What sort of company is wanted, and what sort is already exhausting?

Imagine a person who moves to a new town for work. They accept every invitation, then return home tired and disappointed. At each event they explain again where they have come from and what they do. Nothing has yet lasted long enough for a conversation to begin somewhere else.

They try a neighborhood reading group that meets regularly. It offers a possibility the unrelated events have not: returning to some of the same people. Whether they want to return will depend on what happens in the room.

The difficulty at the doorway

Advice about loneliness often focuses on the moment of approach: introduce yourself, send a message, join a group. It can underestimate everything that makes the approach costly.

A person may have been rejected before. They may be conserving limited energy. They may need transport, an interpreter, an accessible entrance, a place where a child is welcome, or assurance that their identity will not become a subject of debate. Reaching the doorway is already a task.

Once inside, the social arrangement still matters. Are newcomers introduced? Does conversation depend on shared history they cannot know? Can they participate without immediately performing

confidence? Is there a reason to return that does not require inventing a fresh invitation every time?

An established group can decide who welcomes a newcomer, explain the activity, and make the next meeting easy to find. Once the person arrives, someone still has to take an interest. An invitation with a time and place can open the door; remembering the conversation can give them a reason to return.

Inner work and actual company

The interpretation of an interaction can also matter. An unanswered message may be read as rejection even when the other person is distracted. Expecting rejection can make a new encounter difficult to enter openly. But interpretations should be examined in relation to evidence; some groups really do exclude, and some relationships really are neglectful.

A review of randomized studies found average reductions in loneliness from psychological interventions, with considerable variation across studies and results.[51] Help testing a harsh interpretation or tolerating uncertainty could make an encounter easier to enter. The encounter still needs people willing to respond.

A conversation through a screen can offer understanding, continuity, and companionship. It may leave nearby practical needs unmet. Asking what the relationship provides is more revealing than classifying it as online or offline.

For people whose distance from family or others is protective, advice to reconnect can ask them to give up safety.

To be expected somewhere

There is a difference between being allowed to attend and having a place.

The person who always organizes, listens, or cleans up may be needed without feeling known. Reciprocity involves interest in the person as well as appreciation of what they provide.

It also allows variation. A friendship can survive unequal periods of giving when illness, grief, or work makes symmetry impossible. The larger question is whether care can move in more than one direction over time, and whether limits can be spoken without threatening belonging.

At the reading group, our newcomer initially sits at the edge and listens. On a later visit, someone remembers the book they mentioned and asks whether they finished it. The answer starts a conversation they could not have had on the first evening. They no longer have to introduce their whole life before saying something about it.

In time they choose a passage for the group, help set out chairs, and disagree about a book. Someone asks whether they will be there next week. The question is ordinary. Being expected is new.

CHAPTER 11

What makes a life matter

Imagine Jo leaving a retirement party with flowers, thanks, and a box of things from a desk. They are pleased to be leaving. They are also unsure what Monday is for.

For years, the job supplied a schedule, people who expected an answer, problems worth solving, and a familiar way to describe oneself. Some of those things were burdens. Losing the burdens does not prevent missing the structure.

By the second week, the box has been unpacked. There are jobs to do around the house, and some get done. Nobody is waiting for the report. A conversation begins, and the old answer to what one does no longer quite works.

The question is larger than how to stay busy. Busy doing what, with whom, and why?

What has gone missing?

Researchers distinguish several questions that can hide inside the word meaning. Frank Martela and Michael Steger describe coherence, purpose, and significance. Login George and Crystal Park develop a related account using comprehension, purpose, and mattering.[52] The terms separate a life that makes sense, a life with direction, and a life experienced as having value.

Jo has not lost all three. Leaving the job was an understandable decision, and the family’s affection has not vanished. What is missing most sharply is a reason to arrange Tuesday differently from Wednesday, along with the small recognition of being expected somewhere. A new commitment might help with that absence without having to supply a whole new identity.

An invitation arrives to help at a community kitchen. A regular morning shift supplies company and work whose result can be seen

before going home. There is food to prepare, somebody to show a new volunteer where things belong, and a reason to learn names. Jo starts with one morning a week.

The kitchen need not answer every question about the years ahead. For now, there is a shift to turn up for.

The work proves satisfying. Soon there is a second morning, then responsibility for arranging cover. The old skill of keeping an organization running has found a new home. So has the old habit of saying yes when nobody else does.

Being needed has a cost

One week Jo cancels a visit to a friend to cover another absence. The immediate decision is easy enough: the kitchen is short of help. The harder question comes when the same thing happens again.

The work still matters. That is part of the difficulty. Advice to find a purpose would send Jo further into a commitment that is already crowding out other things they care about. The pleasure of being useful has become entangled with the fear of being dispensable.

If Jo stops arranging cover, somebody else has to take it on. There may be fewer meals on a day when no replacement can be found.

Jo tells the coordinator they will keep the original morning shift but stop managing the rota at the end of the month. They spend part of that month showing two other volunteers how it works. The new arrangement remains awkward. A shift goes unfilled. Jo hears about it and feels guilty, without taking the job back.

How much of a finite life can a worthwhile cause reasonably claim? The answer cannot be read from how important the cause is. It must include what the person is being asked to give up.

The answer also depends on what we think people are worth when they cannot contribute. If value rests entirely on usefulness, illness threatens more than capacity. It threatens the person’s claim to care. Rest becomes permissible only when it improves later performance.

The position I take here is that people have claims on our concern that do not depend on their productivity. Research into what people experience as meaningful cannot decide whose life deserves care.

There may be nothing to accomplish

After stepping back from the rota, Jo has an unclaimed afternoon. The first impulse is to fill it. Instead, they put on a piece of music they have not heard for years and stay until it ends.

The afternoon would make a poor line on an application form. Jo was glad to have it.

Researchers have studied experiential appreciation: valuing experiences as they occur. Across seven studies, appreciation was associated with judgments that life was meaningful, beyond the contribution of measures of coherence, purpose, and mattering.[53]

Making every good experience justify itself through a future benefit narrows what we allow ourselves to notice. Jo has listened to music for an afternoon, without putting it to work.

A story can leave things unresolved

Retirement is not the only transition that unsettles an account of a life. Loss, illness, migration, an abandoned ambition, or an unexpected responsibility can make the old story difficult to continue.

The demand for a new story can become another burden. If every loss must have been necessary for growth, a bereaved person is asked to produce the lesson that justifies what happened. If every setback is preparation for success, a life without that success is left without an acceptable ending.

A truthful account can contain accident, waste, injustice, and unanswered questions. Something may have changed you without having been good for you. You may enjoy an afternoon without having finished grieving.

A person can honor someone who died through an ongoing act of care, while continuing to wish the loss had never happened. Finding value afterward gives an observer no right to declare the suffering meaningful. That interpretation belongs first to the person who endured it.

This leaves room for stories to change without demanding that the earlier self be disowned. Jo may once have taken pride in being indispensable. They can recognize what that pride gave them and what it cost. Neither recognition requires pretending the working years were wasted.

A purpose has to answer to other people

Feeling committed does not settle whether a commitment is defensible.

A cause can offer belonging and a reason to endure hardship while directing hostility toward outsiders. It can reward certainty and make doubt feel like betrayal. The strength of the conviction tells us little about whether its account of the world is true or its demands are just.

Commitments need opportunities to encounter correction. Can the people affected describe their costs? Can members question a claim without losing all belonging? Can somebody rest, leave, or change their mind? A purpose that cannot tolerate these questions deserves scrutiny, however meaningful participation feels.

By the following Monday, Jo has neither a grand purpose nor an empty week. There is a shift they intend to keep, a friend to visit, and some time whose use has not been settled. They still miss parts of the job. The new life has begun to acquire a shape of its own.

CHAPTER 12

The stories we tell about one another

Suppose several neighbors share a small garden. They contribute different amounts of time, have different abilities, and do not always agree about what the space is for.

One person seems never to water. Another takes more produce than expected. Somebody explains that the first person has been caring for an ill partner; somebody else disputes the explanation. Soon the group is discussing not only vegetables but character.

A meeting produces an awkward discovery. There was a watering list, but nobody kept it current. The supposed limit on produce was something several members thought everybody understood. Two people have been doing much of the work, and they are tired of correcting the gaps. Others did not know how large the gaps had become.

The group has real grievances and uncertain information. It needs a decision before the next dry week.

What a reputation tells us

We cannot witness everything another person does. Information from others helps us decide whom to trust and what arrangements need attention.

Matthew Feinberg, Robb Willer, and Michael Schultz tested gossip and exclusion in a laboratory public-goods setting. Participants could contribute to a shared fund, and in some conditions information about behavior could reach future partners. When participants could use reputational information to exclude others, the arrangements supported greater cooperation in the task.[54]

The study made contributions visible and available for others to judge. In the garden, the reliability of that information becomes a problem of its own: care elsewhere may be invisible, and a persuasive storyteller may be wrong.

Gossip gives a group a way to share warnings and recognize reliability. It also gives accusations a route to travel. What information is being passed, and what are others being invited to do with it?

There is a considerable distance between reporting that someone missed a task and declaring them selfish. The first claim can be checked. The second proposes a general account of the person. Once accepted, that account can absorb almost anything: another absence confirms it, while a helpful act is dismissed as a performance.

In this garden, the neighbor caring for an ill partner can still answer a specific question about an agreed task. Care responsibilities do not tell us whether a particular promise was made or kept. They do matter when deciding what promise can reasonably be made next.

A group needs room for both kinds of information. Immediate protection may be necessary when harm is credibly reported; fairness does not require leaving people exposed until uncertainty disappears. It does require keeping observation distinct from inference and leaving a route for an account to be corrected.

A clear rule, at last

The gardeners agree that each member will do two hours a week. They put the rota where everybody can see it. Two missed weeks will lead to a review of membership. Produce will be shared at a regular collection time, with a stated limit while supplies are short.

For the people who have been carrying the work, the decision is a relief. Expectations can now be pointed to. A missed task no longer requires a fresh argument about whether it was anybody’s responsibility.

For the caregiver, a fixed block away from home is difficult. Another member can manage watering but cannot do two hours of digging. The equal-hours rule has made one kind of fairness visible while neglecting differences in what the hours require.

The group has reached a conflict between legitimate concerns: dependable work, usable access, and a limit on what anyone can be expected to absorb. The two overworked members have good reason to insist that the gaps be covered by someone else.

Enforcement can introduce trouble of its own. Benedikt Herrmann, Christian Thöni, and Simon Gächter compared public-goods experiments across sixteen participant pools. Some participants punished high contributors, and in some pools this antisocial punishment undermined the cooperation-enhancing effect of punishment.[55] The power to impose a penalty leaves open the question of what behavior will actually attract it.

The garden’s new rule therefore needs more than an enforcement mechanism. Members need a way to say what it requires of them before missing a shift becomes proof that they do not care.

Repair has to mean something

The member who took extra produce acknowledges doing so. But because the old limit was never clearly agreed, the group cannot honestly describe this as the violation of a rule everyone knew. It can explain the shortage, make the new limit explicit, and ask that it be followed.

Suppose the same person takes extra again after agreeing to the limit. The information has changed. There is now a particular promise to discuss and a loss borne by other members. Repair might involve replacing what was taken or accepting restricted access. It need not begin with a verdict about the person’s entire character.

June Tangney, Jeff Stuewig, and Debra Mashek review a distinction between guilt focused on a particular action and shame focused more globally on the self. In the literature they discuss, guilt is often linked with reparative tendencies, while shame can be associated

with concealment or defensiveness.[56] These patterns do not establish whether a feeling is deserved. They do help us ask whether distress is bringing the consequences into view or burying them beneath a judgment of the self.

Declaring oneself a terrible person can sound severe while leaving everyone else’s losses untouched. Conversely, a group can claim to want repair while making repair impossible. If every future contribution will be read as manipulation, what would count as becoming trustworthy again?

Some harms justify lasting limits, and a harmed person does not owe reconciliation. Where changed conduct is the aim, however, the response should make clear what change matters. Otherwise the punishment can continue long after anyone has stopped asking what it is supposed to achieve.

A message about the repeated shortage reaches the neighborhood group. Replies arrive from people who have never seen the rota. The accusation is easier to follow than the history of how the rule was made.

Research combining Twitter observations and behavioral experiments linked social feedback and perceived network norms to later public expressions of moral outrage.[57] In the garden dispute, an applauded accusation might become difficult to revise publicly after a more complicated account emerges.

What fairness asks of the group

Economic-game research across fifteen small-scale societies found substantial variation in behavior, with patterns related to everyday social and economic arrangements.[58] No single familiar setting can supply a human baseline for fairness. The gardeners themselves have several principles available: equal hours, equal benefit, contribution according to capacity, and priority according to need. Agreement that everybody should be fair leaves them with a choice among these principles.

The dispute might survive even if the gardeners learned everything they had asked to know. One of the two members who have been covering gaps understands that the caregiver cannot reliably leave home for a two-hour shift. They still support the equal-hours rule. They fear that every exception will bring another request to cover missed work.

They also want a rule that does not require another meeting whenever a shift is missed. Predictability would let them make plans of their own. The caregiver needs an arrangement they can use. A group cannot settle every future difficulty in advance, but nor can it require each member to keep explaining their life to secure an exception.

The argument I have been making has a limit here. Some members may understand a burden and prefer an arrangement that leaves someone else carrying it. Others may object to a change because it would impose a cost they cannot bear themselves. We cannot decide between these possibilities merely by noticing resistance.

More information can distinguish an impossible duty from an unwanted one, or reveal that a proposed alternative has overlooked necessary work. It cannot establish how much inconvenience one member should accept to preserve another’s access. That requires a judgment about their claims. The people affected also need some influence over the decision; being allowed to explain a burden is little protection if those benefiting from it can simply disregard the answer.

The group might retain a rule, change it, or reduce what it undertakes. A member might refuse an arrangement they consider unfair. Whatever follows, they need to know who will do the work, who will receive the benefit, and what happens to someone who cannot agree.

The group could refuse a trial altogether. Suppose a majority hears the caregiver’s explanation, accepts it as true, and votes to retain the two-hour requirement. The caregiver misses the required shifts and loses membership. The group may now have a clearer rule and fewer arguments. Someone has also lost access to the garden.

Another account of the care responsibilities would add little; the members already believe it. The caregiver might ask allies to seek a new vote, look for another group, or decide that further appeals cost too much. Understanding has located the decision that matters. It has supplied neither the authority nor the agreement to alter it.

Suppose instead that the members agree to examine alternatives. At the next meeting, the group separates the decisions about work and produce. First it lists the jobs that must actually be done. Some need strength, some need brief visits, and some can be done from home. The caregiver offers to maintain the rota and arrange swaps. The members who have been doing most of the watering want to know who will cover a gap if that arrangement fails.

Their question deserves an answer. A proposal to recognize hidden work can still leave visible work undone. The member who favors equal hours accepts a month’s trial on one condition: an unfilled job must remain visible rather than silently becoming theirs. They still prefer the original rule. The group agrees to record unfinished tasks as well as completed ones and review the arrangement after a month. Keeping the record is itself a new job.

The next month will test whether the new duties can actually be carried. An unfinished watering job will require a response, and somebody will have to make it.

Eventually a newcomer will ask how the garden works. Somebody will explain the rota, the limits, and perhaps the argument that produced them. By then the group will have made more than a decision. It will have made something to hand on.

CHAPTER 13

The world we hand on

A day-care center can close at four. That does not make every parent arrive by four.

At ten centers in Haifa, Uri Gneezy and Aldo Rustichini recorded late pickups for four weeks. Six randomly selected centers then introduced a fine of ten shekels for arriving more than ten minutes late. Four continued without it.

Late pickups increased where the fine was imposed. Removing it did not undo the change during the four remaining weeks of observation.[59]

What had the parents been charged for? The authors suggested that the fee might have made lateness look like an extra service with a price. The study did not establish the parents' reasoning or test a larger fine.[59]

A rule enters a world in which people are already interpreting one another. It can change what an action costs and what the action seems to mean. The people who inherit the rule may encounter both changes without knowing how either began.

What we do not have to discover again

An ordinary recipe contains knowledge its reader did not have to discover.

Somebody learned what could be eaten, how ingredients behaved, which sequence worked, and how to communicate enough of the result for another person to begin. The recipe may have changed many times. The cook can change it again.

Reading a timetable requires shared symbols and conventions. Using a medicine depends on knowledge, institutions, and the labor of people one will never meet. Even disagreeing with an inherited

idea requires language and methods acquired from others. Individual ability has a world of accumulated work to act upon.

Alex Mesoudi and Alex Thornton distinguish cumulative cultural evolution from transmission alone. In their account, changes are socially passed on, and further modifications can produce successive improvements in a particular performance. They also emphasize that researchers use the term in different ways and should specify what they are claiming to demonstrate.[60]

A practice can spread because it is familiar, enforced, or useful to influential people. We also need to ask what its improvements serve. A method can become more effective at surveillance without improving the lives of those watched.

A cook can often taste the result. Many cultural consequences are harder to inspect. Benefits can arrive for one group while costs are borne elsewhere. A familiar arrangement may keep working well for the people with the least reason to question it.

The garden's newcomer will inherit more than a rota. Suppose the group welcomes questions at the first meeting, then treats every proposed change as an accusation against the people who did the work before. The newcomer has heard one invitation and encountered another arrangement. Remembering the original dispute could help: the rota itself exists because someone questioned its predecessor.

Shared practices can carry that memory. A review of ritual distinguishes a practice's prescribed sequence from the meaning participants give it.[61] That distinction helps us examine a welcome: who speaks, what is repeated, and what the newcomer is being invited to join. A meal at which an absent member is remembered gives people somewhere to place a loss together. It can also ask someone to repeat an account with which they disagree.

Who prepares the meal, whose history gets told, and whether someone can decline are part of what the next person learns. The value expressed by a gathering eventually meets the way it is conducted.

After the meeting

A month has passed in our imagined garden. The caregiver arranges swaps from home, and the member who struggled with digging takes shorter watering visits. Fewer jobs go unfilled.

One cost becomes apparent only when somebody asks about the record. Maintaining it, answering messages, and finding replacements have taken much longer than expected. The group has made room for different contributions but still failed to count one of them properly.

If it simply congratulates itself on becoming fairer, the hidden work will remain hidden. Instead, the members set a limit on the coordination role and take turns doing it. At the end of the trial, they also reduce the number of beds they cultivate. There will be less produce. There is no longer quite so much unpaid rescue work required to keep the promise.

The problem extends beyond our invented garden. In Nepal, Elinor Ostrom and her colleagues studied irrigation systems in which farmers depended on shared channels to bring water to their fields. The water had to reach people near the end of the system as well as those nearer its source. Someone also had to maintain the channels and ensure that agreed shares were respected.

Ostrom described research by Wai Fung Lam comparing the physical condition of systems, water reaching farms at the tail end, and agricultural productivity. Farmer-governed systems performed better on those measures after accounting for measured environmental differences. Their users met, negotiated agreements, appointed monitors, and sanctioned breaches. These were maintained institutions.[62]

In a later account with Xavier Basurto, Ostrom described arrangements in which farmers took turns serving as guards. Monitoring both water use and maintenance contributions was work performed by users of the system. The rules varied considerably, and farmers in newer systems described having to try combinations

and adjust them over time. In older systems, people could inherit rules whose origins they no longer knew.[63]

These field comparisons cannot isolate the effect of each rule, and farmer-managed systems varied in performance. They bring the work of governing a resource into view: monitoring, maintaining, negotiating, and revising arrangements.

In the garden, the group still needs to ask who can challenge a record and get it corrected. The answer will reveal something about where power rests.

The gardeners' first equal-hours rule made obligations clear. Their second arrangement made different capacities easier to accommodate. Neither ended the work of judgment. Who benefits from the reduced harvest? Who can attend the review meeting? Who is believed when they say a task takes longer than the record shows?

Those questions belong inside the institution, where people can answer them and change a decision. If they exist only in private complaints, the public rule may look successful while the costs accumulate elsewhere.

Whose inheritance?

An institution can look different from each position within it. That is easy to lose when its rules are described as a shared culture.

Culture is often described as though everyone in a country or community shares one mind. The phrase *our culture* can conceal an argument about whose practices count and who gets to name the tradition.

Research can repeat that mistake. An influential critique drew attention to the heavy reliance of behavioral science on Western, educated, industrialized, rich, and democratic populations.[64] Evidence from a narrow sample should not quietly become a claim about all people. Nor does finding a difference between groups make either group internally uniform.

Within a family, workplace, religious community, or nation, people inherit different positions. They can disagree about a practice while appealing to a shared value. A younger member may ask to change a tradition precisely because they believe its promise should include someone previously left out.

Cultural participation therefore involves more than a choice between accepting an inheritance and discarding it. What is worth keeping? Whose experience has been omitted? Which form no longer serves the value it claims to express? Sometimes the answer is preservation. Sometimes refusal. The history cannot make that decision for us, but understanding it can clarify what is at stake.

What the next person meets

A newcomer to the garden will find fewer beds, a rota, and people who remember why the arrangements changed. If the explanation survives, the newcomer can learn from the disagreement as well as from the rule. If only the rule survives, it may eventually look arbitrary.

Whoever enters a room tomorrow will encounter choices made before they arrive. Someone has set the schedule, placed the furniture, chosen which mistakes to record, and established how disagreement is received. Some decisions were deliberate. Others accumulated without anyone considering them together.

Our power over those conditions differs enormously. A manager, planner, parent, participant, and newcomer do not possess the same authority or resources. Responsibility should follow those differences. Wherever influence exists, though, there is a question to ask: what does this arrangement make people carry, and what does it make possible?

The garden leaves its next members a practice, a memory of what it cost, and a way to reconsider it.

CONCLUSION

A different next move

The manager's message from the Introduction is still on the phone: *Can we talk tomorrow?*

At the beginning, we watched the person receiving it. Four words could meet a secure employee, a precarious worker, someone expecting good news, or someone whose recent experience made ambiguity difficult to bear. The sentence alone could not explain the response.

Now consider the sender. Perhaps the manager is hurried. Perhaps the subject needs privacy. Perhaps this workplace has repeatedly made an unexplained meeting the prelude to bad news. Each possibility suggests a different question. None can be settled by looking at the recipient's anxiety alone.

The manager may be able to clarify the subject with one more sentence. The employee may be able to ask. If there is a real threat to the job, however, clarification will not make it disappear. The useful next step may concern notice, support, or a plan for what follows.

An inquiry to take with you

The job applicant in Chapter 1 learned that part of an apparent confidence problem was a conflict over time. The household in the sleep chapter found that a device rule could not do all the work assigned to it. The gardeners had to revise both the distribution of tasks and the amount of work they attempted. In each case, a more specific explanation changed what could be tried.

For an ordinary difficulty in your own life, the inquiry can be kept brief. These questions are prompts for judgment, not a validated procedure.

What happened? Describe one event closely enough that somebody else could recognize it. A missed appointment is an event. Being hopeless is a verdict that leaves almost everything unexplained.

What are the plausible explanations? Separate what you know from what you infer. Include possibilities outside the person's character: an unclear agreement, illness, missing resources, a learned expectation, or a conflict between things that matter. Ask the person rather than inventing their history.

What could distinguish the explanations? Look for information that could change your mind. If the task is unclear, another hour alone with it may not help. An answer from someone else might.

What can change, and who can change it? Choose a feasible next step and include its costs. A quieter evening may require somebody else to take a shift. A fairer rule may require the group to promise less. If you lack the authority or resources, identifying who has them is part of the inquiry. They may understand the problem and still refuse the change; then the question includes how a decision can be challenged and what options remain.

What happened afterward? Notice burdens as well as benefits, using the outcome you chose to watch. Ask what else changed before assigning the result to your intervention. If it fails, reconsider the explanation before turning the failure into another accusation.

Some difficulties need protection, clinical assessment, or material help before they need an informal experiment. Others remain partly unexplained even after careful inquiry. We can acknowledge that uncertainty while taking necessary action.

Responsibility becomes more useful when it is specific. What influence was available? What harm occurred? What protection, repair, or prevention is now possible? An account of why someone broke a promise can help answer those questions. The person who bore the cost still has a claim on what happens next.

There is a limit to how long an inquiry should continue. A person need not keep disclosing painful details to an audience that already understands and refuses to act. Nor should every recurring arrangement require a fresh negotiation. Sometimes the next move is to make an agreement dependable enough that people can get on with their lives. Sometimes it is to challenge a decision, seek support elsewhere, or recognize a loss that greater understanding cannot repair.

The next person

There is no final state in which the human animal becomes perfectly adjusted. Bodies age. Relationships alter. Circumstances change. What once protected someone may begin to restrict them; a possibility once out of reach may become usable with different support.

Our explanations need to remain capable of changing too.

Sometimes that work produces a better private decision. Sometimes it reveals a conversation that needs to happen or a condition that should not be endured alone. Its consequence may also be what another person encounters through us.

Perhaps the manager adds the missing context and the evening becomes easier. Perhaps the message announces a change the employee cannot accept. Either way, there is now something more definite to face than an unexplained sentence. A clearer account can offer relief, reveal a choice, or make a disagreement harder to ignore. The next move still has to be made.

We are creatures of context. We are also, unevenly and together, among its makers.

Notes

Numbers link back to their first occurrence in the text. Source links open the cited material.

Introduction: The person in the room

1. Desmond Morris, *The Human Animal: A Personal View of the Human Species* (Crown, 1994). Bibliographic record: https://search.worldcat.org/title/31170486. This acknowledgment does not claim to summarize or rebut Morris's full argument. Return

Chapter 1: What matters right now

2. The opening questions are the book's synthesis. They draw on overlapping constructs working at different levels, rather than forming a validated taxonomy or anatomical model. Sources appear in Chapters 2, 3, and 11. Return

3. Joseph E. LeDoux and Daniel S. Pine, "Using Neuroscience to Help Understand Fear and Anxiety: A Two-System Framework," *American Journal of Psychiatry* (2016), https://pubmed.ncbi.nlm.nih.gov/27609244/. For an explicit challenge, see Michael S. Fanselow and Zachary T. Pennington, "A Return to the Psychiatric Dark Ages with a Two-System Framework for Fear," *Behaviour Research and Therapy* (2018), https://pubmed.ncbi.nlm.nih.gov/29128585/. This book does not adjudicate that debate or infer subjective experience from a defensive response. Return

Chapter 2: What stress hides

4. Sally S. Dickerson and Margaret E. Kemeny, "Acute Stressors and Cortisol Responses: A Theoretical Integration and Synthesis of Laboratory Research," *Psychological Bulletin* 130, no. 3 (2004): 355-391, https://doi.org/10.1037/0033-2909.130.3.355. The meta-analysis covered 208 healthy-adult laboratory studies. Social-evaluative threat and limited control were associated with larger cortisol responses, especially when combined; the features were partly entangled with difficulty, effort, and unpredictability. Return

5. Idy S. C. Man et al., "Multi-systemic Evaluation of Biological and Emotional Responses to the Trier Social Stress Test: A Meta-analysis and Systematic Review," *Frontiers in Neuroendocrinology* 68 (2023): 101050, https://doi.org/10.1016/j.yfrne.2022.101050. The review covered standard Trier-test studies in healthy adults. Six studies reported correlations between changes in biomarkers and mood; their results were inconsistent. See section 3.5

of the full paper: https://ira.lib.polyu.edu.hk/bitstream/10397/102286/1/1-s2.0-S0091302222000735-main.pdf. Return

6. Jeannette Weber, Peter Angerer, and Jennifer Apolinário-Hagen, “Physiological Reactions to Acute Stressors and Subjective Stress during Daily Life,” *PLOS ONE* 17 (2022): e0271996, https://doi.org/10.1371/journal.pone.0271996. Direct positive links appeared in roughly half of eligible analyses for cortisol, blood pressure, and heart rate. The studies were too heterogeneous and confounded for one pooled estimate. Return

7. Bruce S. McEwen, “Physiology and Neurobiology of Stress and Adaptation: Central Role of the Brain,” *Physiological Reviews* 87 (2007): 873-904, https://doi.org/10.1152/physrev.00041.2006. Full text: https://njc.rockefeller.edu/pdf2/McEwenPhysioRev07.pdf, sections I and III.A-B. This review combines human and animal evidence. Later reviews appear in the stress-and-health note; no individual damage score is inferred. Return

8. James P. Herman and colleagues, “Regulation of the Hypothalamic-Pituitary-Adrenocortical Stress Response,” *Comprehensive Physiology* 6 (2016): 603-621, https://doi.org/10.1002/cphy.c150015. Full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC4867107/. The introduction and sections on HPA-axis activity and glucocorticoid feedback support the account of energy mobilization and negative feedback. Much of the mechanistic research reviewed is in rodents; the chapter uses the broad mammalian organization of the system. Return

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wellbeing, and less exhaustion, but most evidence was self-reported and correlational. Return

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Chapter 4: Learning what to expect

19. Celeste Kidd, Holly Palmeri, and Richard N. Aslin, “Rational Snacking: Young Children’s Decision-Making on the Marshmallow Task Is Moderated by Beliefs about Environmental Reliability,” *Cognition* 126 (2013), https://doi.org/10.1016/j.cognition.2012.08.004. Author-hosted full text: https://www.kiddlab.com/_files/ugd/0975fd_d7e58cfd98604428945ba9afd6d41f64.pdf. The procedure and random assignment are in section 2; Figure 1 reports mean waits in the 28-child experiment. In Lillie Moffett, Carol Flannagan, and Priti Shah, “The Influence of Environmental Reliability in the Marshmallow Task: An Extension Study,” *Journal of Experimental Child Psychology* 194 (2020): 104821, https://doi.org/10.1016/j.jecp.2020.104821, sixty children completed a similar procedure in familiar school settings. Mean waits were about 9.6 versus 7 minutes. The preregistered tests did not reach statistical significance; a Bayesian analysis combining these data with the original study supported an effect. That combined result is not an independent confirmation of the original effect’s size. Full text: https://files.eric.ed.gov/fulltext/ED662939.pdf. Return

20. Tyler W. Watts, Greg J. Duncan, and Haonan Quan, “Revisiting the Marshmallow Test,” *Psychological Science* 29 (2018), https://doi.org/10.1177/0956797618761661. Full text: https://pmc.ncbi.nlm.nih.gov/articles/PMC6050075/. The sample was more diverse than the original Stanford preschool sample, but not nationally representative. Return

21. Laura E. Michaelson and Yuko Munakata, “Same Data Set, Different Conclusions: Preschool Delay of Gratification Predicts Later Behavioral Outcomes in a Preregistered Study,” *Psychological Science* 31, no. 2 (2020): 193-201, https://doi.org/10.1177/0956797619896270. For a related debate about statistical adjustment and what the task measures, see Tyler W. Watts and Greg J. Duncan, “Controlling, Confounding, and Construct Clarity: Responding to Criticisms of ‘Revisiting the Marshmallow Test’ by Doebel, Michaelson, and Munakata (2020) and Falk, Kosse, and Pinger (2020),” *Psychological Science* 31, no. 1 (2020): 105-108, https://doi.org/10.1177/0956797619893606. The latter replies to the commentaries named in its title, not to the Michaelson-Munakata article. Michaelson and Munakata's full text, https://pmc.ncbi.nlm.nih.gov/articles/PMC7197218/, reports the behavioral association after adjustment for preschool behavior in Table 3; their outcome definitions and controls differ from Watts and colleagues'. These are analyses of one cohort, not independent replications. Return

22. Katie A. McLaughlin, Margaret A. Sheridan, and Hilary K. Lambert, “Childhood Adversity and Neural Development: Deprivation and Threat as Distinct Dimensions of Early Experience,” *Neuroscience and Biobehavioral Reviews* 47 (2014), https://doi.org/10.1016/j.neubiorev.2014.10.012. Full text: https://sdlab.fas.harvard.edu/file_url/569, sections 1-3 and 4.1. The proposed neural pathway from social deprivation draws on animal and human evidence; it is not direct observation of synaptic changes in the children discussed here. Return

23. Seth D. Pollak and Pawan Sinha, “Effects of Early Experience on Children’s Recognition of Facial Displays of Emotion,” *Developmental Psychology* 38 (2002): 784-791. Full text: https://web.mit.edu/sinhalab/Papers/effects_early_experience_devpsych.pdf. Methods and results compare 24 physically abused children with 23 comparison children. Exposure groups were not randomized. Recognition of posed facial expressions does not directly measure a child’s feelings during a confrontation. Return

24. Nathan A. Fox and colleagues, “The Effects of Severe Psychosocial Deprivation and Foster Care Intervention on Cognitive Development at 8 Years of Age,” *Journal of Child Psychology and Psychiatry* (2011), https://pubmed.ncbi.nlm.nih.gov/21244422/. For a later synthesis of the same trial, see Lucy S. King and colleagues, “A Comprehensive Multilevel Analysis of the Bucharest Early Intervention Project,” *American Journal of Psychiatry* 180 (2023), https://doi.org/10.1176/appi.ajp.20220672. Repeated reports from this project are follow-ups of one trial, not independent replications. Return

Chapter 5: Averages do not live your life

25. William Fleeson, “Toward a Structure- and Process-Integrated View of Personality: Traits as Density Distributions of States,” *Journal of Personality and Social Psychology* (2001), https://pubmed.ncbi.nlm.nih.gov/11414368/. Full paper: https://personality-project.org/revelle/syllabi/classreadings/fleeson.2001.pdf. Stability refers to the periods studied. The overlapping records are illustrative, not additional observations or evidence of their causes. Return

26. Tinca J. C. Polderman and colleagues, “Meta-analysis of the Heritability of Human Traits Based on Fifty Years of Twin Studies,” *Nature Genetics* 47 (2015), https://doi.org/10.1038/ng.3285. Primary institutional record: https://research.vu.nl/en/publications/meta-analysis-of-the-heritability-of-human-traits-based-on-fifty-/. Twin estimates depend on assumptions about genetic and environmental resemblance and on trait measurement. Broad pooled estimates should not be assigned to an individual’s personality. Return

27. MedlinePlus Genetics, “What Is Heritability?” https://medlineplus.gov/genetics/understanding/inheritance/heritability/; and “Is Height Determined by Genetics?” https://medlineplus.gov/genetics/understanding/traits/height/. These explain population dependence and the roles of genes and environment. The nutrition example illustrates the difference between explaining variation under given conditions and changing those conditions; it assigns no numerical heritability estimate. Return

28. Michael Pluess, “Individual Differences in Environmental Sensitivity,” *Child Development Perspectives* 9 (2015), https://doi.org/10.1111/cdep.12120. This is an integrative conceptual framework, not a validated system for classifying every individual. Return

29. Brent W. Roberts and colleagues, “A Systematic Review of Personality Trait Change through Intervention,” *Psychological Bulletin* (2017), https://doi.org/10.1037/bul0000088. Primary institutional record: https://experts.illinois.edu/en/publications/a-systematic-review-of-personality-trait-change-through-intervent/. Trait measurement, symptom change, study design, and intervention context complicate interpretation. Return

Chapter 6: A place for sleep

30. Alexander A. Borbély, Serge Daan, Anna Wirz-Justice, and Tom Deboer, “The Two-Process Model of Sleep Regulation: A Reappraisal,” *Journal of Sleep Research* 25, no. 2 (2016): 131-143. https://doi.org/10.1111/jsr.12371. Return

31. Anne-Marie Chang, Daniel Aeschbach, Jeanne F. Duffy, and Charles A. Czeisler, “Evening Use of Light-Emitting eReaders Negatively Affects Sleep, Circadian Timing, and Next-Morning Alertness,” *Proceedings of the National*

Academy of Sciences 112, no. 4 (2015): 1232-1237. https://doi.org/10.1073/pnas.1418490112. Return

32. Hans P. A. Van Dongen, Greg Maislin, Janet M. Mullington, and David F. Dinges, "The Cumulative Cost of Additional Wakefulness: Dose-Response Effects on Neurobehavioral Functions and Sleep Physiology from Chronic Sleep Restriction and Total Sleep Deprivation," *Sleep* 26, no. 2 (2003): 117-126. https://doi.org/10.1093/sleep/26.2.117. Return

33. Jack D. Edinger, J. Todd Arnedt, Suzanne M. Bertisch, et al., "Behavioral and Psychological Treatments for Chronic Insomnia Disorder in Adults: An American Academy of Sleep Medicine Clinical Practice Guideline," *Journal of Clinical Sleep Medicine* 17, no. 2 (2021): 255-262. https://doi.org/10.5664/jcsm.8986. Return

34. Daniel J. Buysse, J. Todd Arnedt, Luis Buenaver, et al., "Combination Treatment for Chronic Insomnia Disorder in Adults: An American Academy of Sleep Medicine Clinical Practice Guideline," *Journal of Clinical Sleep Medicine* 22 (2026), article 56. https://doi.org/10.1007/s44470-025-00038-8. The introduction explicitly discusses clinical history, diagnosis, and co-occurring conditions. Its treatment recommendations concern adding medication to CBT-I; this chapter does not attempt to guide medication choices. Return

Chapter 7: Room to move

35. Ulf Ekelund and colleagues, "Dose-Response Associations between Accelerometry Measured Physical Activity and Sedentary Time and All Cause Mortality," *BMJ* 366 (2019), https://doi.org/10.1136/bmj.l4570. These are observational cohort data, not randomly assigned activity exposures. Return

36. World Health Organization, *WHO Guidelines on Physical Activity and Sedentary Behaviour* (2020), https://www.who.int/publications/i/item/9789240015128; and "Physical Activity," https://www.who.int/news-room/fact-sheets/detail/physical-activity. Population guidance is not an individualized exercise prescription. Return

37. Michael Noetel and colleagues, "Effect of Exercise for Depression," *BMJ* 384 (2024), https://doi.org/10.1136/bmj-2023-075847; correction, https://doi.org/10.1136/bmj.q1024. The review reported substantial bias concerns and low or very low confidence for important comparisons. No exact effect size or universal ranking is used here. Return

38. Andrew J. Clegg and colleagues, "Exercise for Depression," *Cochrane Database of Systematic Reviews* 1 (2026): CD004366, https://doi.org/10.1002/14651858.CD004366.pub7; indexed abstract: https://pubmed.ncbi.nlm.nih.gov/41500513/. Searches ran through November 2023. This review and Noetel and colleagues' review include some of the same trials; agreement partly reflects re-examining the same participants' results. The

wording here follows the updated review's cautious conclusion and does not claim equivalence to medication or psychotherapy. Return

39. National Institute for Health and Care Excellence, *ME/CFS: Diagnosis and Management*, guideline NG206 (2021), recommendations 1.11.9-1.11.14, https://www.nice.org.uk/guidance/ng206/chapter/Recommendations. This boundary is specific to ME/CFS; other conditions also require appropriate individual assessment. Return

Chapter 8: What to do with a feeling

40. James J. Gross, "Emotion Regulation: Current Status and Future Prospects," *Psychological Inquiry* 26 (2015), https://doi.org/10.1080/1047840X.2014.940781. The process models distinguish possible targets and stages of regulation; the everyday applications in this chapter are illustrative reasoning, not a tested treatment protocol. Return

41. Allison S. Troy, Amanda J. Shallcross, and Iris B. Mauss, "A Person-by-Situation Approach to Emotion Regulation: Cognitive Reappraisal Can Either Help or Hurt, Depending on the Context," *Psychological Science* 24 (2013): 2505-2514, https://doi.org/10.1177/0956797613496434; corrigendum (2016), https://doi.org/10.1177/0956797615627417. The correction amended seven depression scores; the authors reported unchanged substantive conclusions. The body describes an association at high stress severity, not an experimental effect of prescribing reappraisal. Return

42. Brett Q. Ford, Phoebe Lam, Oliver P. John, and Iris B. Mauss, "The Psychological Health Benefits of Accepting Negative Emotions and Thoughts," *Journal of Personality and Social Psychology* 115 (2018; first published online 2017), https://doi.org/10.1037/pspp0000157. The laboratory, diary, and longitudinal findings concern patterns of acceptance and health; they do not establish a universal immediate intervention effect. Return

43. Emily A. Butler, Tiane L. Lee, and James J. Gross, "Emotion Regulation and Culture: Are the Social Consequences of Emotion Suppression Culture-Specific?" *Emotion* 7 (2007), https://doi.org/10.1037/1528-3542.7.1.30. Cultural values were measured within the study; the result should not be converted into fixed rules for ethnic groups. Return

Chapter 9: Beginning, repeating, avoiding

44. Piers Steel, "The Nature of Procrastination," *Psychological Bulletin* 133 (2007), https://doi.org/10.1037/0033-2909.133.1.65. The definition distinguishes dysfunctional postponement from delay in general; the meta-analysis and theoretical account identify multiple relevant factors rather than a single cause. Return

45. Fuschia M. Sirois and Timothy A. Pychyl, “Procrastination and the Priority of Short-Term Mood Regulation: Consequences for Future Self,” *Social and Personality Psychology Compass* 7 (2013), https://doi.org/10.1111/spc3.12011. This is a theoretical synthesis emphasizing mood repair and consequences over time, not proof that every instance of delay has the same mechanism. Return

46. Wendy Wood and Dennis Rünger, “Psychology of Habit,” *Annual Review of Psychology* 67 (2016), https://doi.org/10.1146/annurev-psych-122414-033417. Repetition, habitual control, and deliberate intention are related but not interchangeable. Return

47. Phillippa Lally, Cornelia H. M. van Jaarsveld, Henry W. W. Potts, and Jane Wardle, “How Are Habits Formed: Modelling Habit Formation in the Real World,” *European Journal of Social Psychology* 40 (2010; online 2009), https://doi.org/10.1002/ejsp.674. The 18-254-day range estimates time to 95% of each fitted plateau among 39 participants whose curves met the authors' good-fit criteria (Table 2). Records covered 84 days, so some estimates extrapolate beyond observation. Automaticity was self-reported. The publisher's abstract and full paper were checked. Return

48. Peter M. Gollwitzer and Paschal Sheeran, “Implementation Intentions and Goal Achievement,” *Advances in Experimental Social Psychology* 38 (2006), https://doi.org/10.1016/S0065-2601(06)38002-1. The illustrative plans and design questions here are applications, not a replication of a particular intervention. Return

49. Alexander Rozental and colleagues, “Targeting Procrastination Using Psychological Treatments,” *Frontiers in Psychology* 9 (2018), https://doi.org/10.3389/fpsyg.2018.01588. Outcomes were self-reported, comparisons were with inactive controls, and the evidence was heterogeneous; stronger conclusions about particular treatments need more evidence. Return

Chapter 10: A place among others

50. Julianne Holt-Lunstad, Timothy B. Smith, Mark Baker, Tyler Harris, and David Stephenson, “Loneliness and Social Isolation as Risk Factors for Mortality,” *Perspectives on Psychological Science* 10 (2015), https://doi.org/10.1177/1745691614568352; Fan Wang and colleagues, “A Systematic Review and Meta-Analysis of 90 Cohort Studies of Social Isolation, Loneliness and Mortality,” *Nature Human Behaviour* 7 (2023), https://doi.org/10.1038/s41562-023-01617-6. These syntheses concern associations in observational studies, with overlap in the underlying literature. Loneliness and isolation are related but distinct constructs. Return

51. Nisha Hickin and colleagues, “The Effectiveness of Psychological Interventions for Loneliness,” *Clinical Psychology Review* 88 (2021), https://doi.org/10.1016/j.cpr.2021.102066. The review found average benefits alongside substantial variation and limitations in study quality. The group-design

suggestions in this chapter are reasoned applications, not the intervention tested by this review. Return

Chapter 11: What makes a life matter

52. Frank Martela and Michael F. Steger, "The Three Meanings of Meaning in Life: Distinguishing Coherence, Purpose, and Significance," *The Journal of Positive Psychology* 11 (2016): 531-545, https://doi.org/10.1080/17439760.2015.1137623; Login S. George and Crystal L. Park, "Meaning in Life as Comprehension, Purpose, and Mattering," *Review of General Psychology* 20 (2016), https://doi.org/10.1037/gpr0000077. These are conceptual distinctions within an overlapping literature. The retirement and kitchen sequence is invented. Claims about worth, obligation, grief, and defensible commitments are ethical arguments, not effects established by these papers. Return

53. Jinhyung Kim and colleagues, "Experiential Appreciation as a Pathway to Meaning in Life," *Nature Human Behaviour* 6 (2022): 677-690, https://doi.org/10.1038/s41562-021-01283-6. Author-uploaded full text: https://www.researchgate.net/publication/358516348_Experiential_appreciation_as_a_pathway_to_meaning_in_life. The seven studies combine surveys, daily diaries, and experiments. The chapter describes the association with meaning judgments after accounting for coherence, purpose, and mattering. The authors report relatively weak experimental effects and questions about generalizability; the finding is not treated here as a clinical intervention or a complete theory of meaning. Return

Chapter 12: The stories we tell about one another

54. Matthew Feinberg, Robb Willer, and Michael Schultz, "Gossip and Ostracism Promote Cooperation in Groups," *Psychological Science* 25 (2014), https://doi.org/10.1177/0956797613510184. The laboratory study used structured public-goods decisions and specified channels for information and partner exclusion. The garden is an invented illustration throughout, not a field case from this study. Return

55. Benedikt Herrmann, Christian Thöni, and Simon Gächter, "Antisocial Punishment Across Societies," *Science* 319, no. 5868 (2008): 1362-1367, https://doi.org/10.1126/science.1153808. Full text: https://www.umass.edu/preferen/You%20Must%20Read%20This/herrmann-thoni-gachter.pdf. The experiments involved 1,120 university students in sixteen participant pools, playing repeated public-goods games with real monetary stakes. Antisocial punishment meant penalizing someone who contributed at least as much as the punisher. The pools are not representative national samples; the chapter makes no inference about national character. Return

56. June Price Tangney, Jeff Stuewig, and Debra J. Mashek, "Moral Emotions and Moral Behavior," *Annual Review of Psychology* 58 (2007),

https://doi.org/10.1146/annurev.psych.56.091103.070145. The self-versus-behavior distinction describes tendencies within a varied literature; moral emotions do not independently establish moral truth. Return

57. William J. Brady, Killian McLoughlin, Tuan N. Doan, and Molly J. Crockett, "How Social Learning Amplifies Moral Outrage Expression in Online Social Networks," *Science Advances* 7 (2021), https://doi.org/10.1126/sciadv.abe5641. Observed platform behavior and experimental expression are not direct measurements of private felt outrage, and the findings should not be generalized to every platform feature. Return

58. Joseph Henrich and colleagues, "'Economic Man' in Cross-Cultural Perspective: Behavioral Experiments in 15 Small-Scale Societies," *Behavioral and Brain Sciences* 28 (2005), https://pubmed.ncbi.nlm.nih.gov/16372952/. The comparisons reveal variation and associations with social conditions, not a simple experimental cause of cultural differences. Return

Chapter 13: The world we hand on

59. Uri Gneezy and Aldo Rustichini, "A Fine Is a Price," *The Journal of Legal Studies* 29, no. 1 (2000): 1-17, https://doi.org/10.1086/468061. The published methods and results were checked. The authors initially recruited twelve centers; two were excluded because records were incomplete. For a later test using a different method, see Cherie Metcalf, Emily Satterthwaite, J. Shahar Dillbary, and Brock Stoddard, "Is a Fine Still a Price? Replication as Robustness in Empirical Legal Studies," *International Review of Law and Economics* 63 (2020): 105906, https://doi.org/10.1016/j.irle.2020.105906. Author-uploaded full text: https://www.researchgate.net/publication/342615363_Is_a_fine_still_a_price_Replication_as_robustness_in_empirical_legal_studies, sections II-III. Their randomized online vignette studies found reductions in intended lateness duration and tax noncompliance when fines were introduced, although not every outcome changed significantly. Hypothetical responses are not actual daycare pickups, so this tests robustness across methods and settings rather than directly refuting the field result. Return

60. Alex Mesoudi and Alex Thornton, "What Is Cumulative Cultural Evolution?" *Proceedings of the Royal Society B* 285 (2018), https://doi.org/10.1098/rspb.2018.0712. The authors distinguish core and extended criteria and caution against treating the concept as a single boundary between human and nonhuman species. Cultural transmission does not imply automatic moral progress. Return

61. Nicholas M. Hobson, Juliana Schroeder, Jane L. Risen, Dimitris Xygalatas, and Michael Inzlicht, "The Psychology of Rituals," *Personality and Social Psychology Review* 22 (2018), https://doi.org/10.1177/1088868317734944. The review distinguishes physical features of ritual from meaning and offers an integrative framework. The meal and welcome are illustrations, not reported experiments. Return

62. Elinor Ostrom, “Beyond Markets and States: Polycentric Governance of Complex Economic Systems,” *American Economic Review* 100 (2010), https://doi.org/10.1257/aer.100.3.641. Full text: https://web.pdx.edu/~nwallace/EHP/OstromPolyGov.pdf, section VI.A, pp. 656-657. The Nepal comparison is reported from Ostrom's account; Lam's 1998 book was not independently inspected. The garden is illustrative, not field evidence. Return

63. Elinor Ostrom and Xavier Basurto, “Crafting Analytical Tools to Study Institutional Change,” *Journal of Institutional Economics* 7 (2011; online 2010): 317-343, https://doi.org/10.1017/S1744137410000305. Section 2 reports rotating guards, varied rules, and accounts of their origins and revision. This draws on the same research program as the preceding comparison, not an independent replication. Return

64. Joseph Henrich, Steven J. Heine, and Ara Norenzayan, “The Weirdest People in the World?” *Behavioral and Brain Sciences* 33 (2010), https://doi.org/10.1017/S0140525X0999152X. The sampling warning applies to this book’s generalizations and examples too. Return